TUNIS BUSINESS SCHOOL

Chapter 1

Job-Shop Scheduling



Static scheduling models — from one machine to m machines

Production & Operations Management

Dr Miryam Daoud Marrakchi

October 2026

CHAPTER CONTENTS

What we will cover

Scheduling n jobs on one machine	Priority rules · MOT rule · Gantt chart
Scheduling n jobs on two machines	Johnson's algorithm · Jackson's algorithm
Scheduling n jobs on three machines	Dominance condition · reduction to two machines
Scheduling n jobs on m machines	Heuristic method · large problems
Exercise after every section + final review	Current-life data · detailed solutions



CHAPTER OBJECTIVE

Chapter objective

Scheduling problems consist in determining the optimal order in which a given number of jobs (tasks) must be executed over a given period. Completing a job requires passing through one or several machines.

> ***We look for the sequence that optimises a chosen criterion — the mean completion time (one machine) or the total execution time, the makespan (several machines).***

INTRODUCTION

Job-shops and scheduling models

This chapter deals with batch production in job-shops. A job-shop is organised by function: all the equipment performing the same technical function is gathered in the same place, and jobs travel between these areas.

STATIC models

All jobs are known in advance and no new job enters the schedule during the period considered.

DYNAMIC models

Jobs arrive successively over time, most often in a random environment.

This chapter studies static models: 1 machine, then 2, then 3, and finally m machines.

NEXT → *We start with the simplest case — n jobs on ONE machine.*

1. SCHEDULING n JOBS ON ONE MACHINE

Example — five jobs on machine A

Five jobs must be processed on a single machine A. Let Ti be the operating time of job i. All times are in minutes.

Job i	1	2	3	4	5	Total
Ti (min)	50	150	80	200	30	**510**

QUESTIONS

1. Determine the optimal sequence of the jobs.
2. Draw the Gantt chart.
3. Determine the minimum mean completion time.

> ***Whatever the order, the machine works 510 min. Another criterion is needed.***

1.1 PRIORITY RULES ON ONE MACHINE

Several rules exist

Rule	How the next job is chosen	Main objective
FIFO / FCFS	First In, First Out — jobs in order of arrival	Fairness, simplicity
SPT = MOT	Shortest (Minimum) Operating Time first	Minimise mean completion time
LPT	Longest Processing Time first	Balance load between machines
EDD	Earliest Due Date first	Minimise maximum lateness
CR	Critical Ratio = time to due date / processing time	Urgency with deadlines

Each rule serves a different objective: the choice of rule depends on what we want to optimise.

1.1 THE MOT RULE (MINIMUM OPERATING TIME)

Why we retain the MOT rule

Optimal

It gives the minimum mean completion time — proven: no other sequence does better.

Less waiting

Jobs leave the shop earlier on average: less waiting time and less work-in-progress.

Simple

Sort the operating times in increasing order — nothing more.

DEFINITION **The MOT rule processes the jobs in increasing order of operating time: $T1 \leq T2 \leq \ldots \leq Tn$.**

Limit: MOT ignores due dates — a long job may wait a long time. When deadlines matter, EDD is preferred.

1.1 THE MOT RULE

Notation: completion time Cj

Tj	Operating time of the job placed in position j of the sequence.
Cj	Completion time of the job in position j: its own time plus the times of all preceding jobs.

COMPLETION TIME

$$Cj = T1 + T2 + \ldots + Tj$$

MEAN COMPLETION TIME

$$\bar{C} = (1/n) \sum j=1..n \; Cj$$

Objective of the MOT rule: make $\bar{C}$ as small as possible.

1.1 EXAMPLE — SOLUTION

Applying the MOT rule

Step 1 — sort the jobs by increasing operating time. Step 2 — compute Cj. Step 3 — compute C.

Position j	1	2	3	4	5	Total
Job (MOT order)	5	1	3	2	4	—
Tj (min)	30	50	80	150	200	**510**
Cj (min)	30	80	160	310	510	**1 090**

Rule	Sequence	C (min)
MOT / SPT	**5 – 1 – 3 – 2 – 4**	**218**
FIFO	1 – 2 – 3 – 4 – 5	304
LPT	4 – 2 – 3 – 1 – 5	394

MINIMUM MEAN COMPLETION TIME

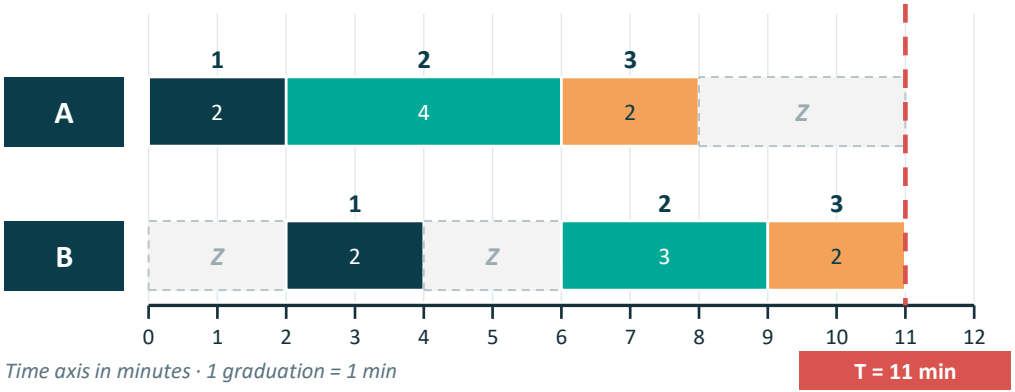
$\bar{C} = 1090 / 5 = 218$ min

1.2 THE GANTT CHART

How to read and draw a Gantt chart

- Time runs from left to right, starting at 0.
- One line per machine; one bar per job — length = operating time.
- The job number is written above its bar.
- Z = idle time: the machine waits.
- **A job starts on a machine only when it has left the previous machine AND the machine is free.**

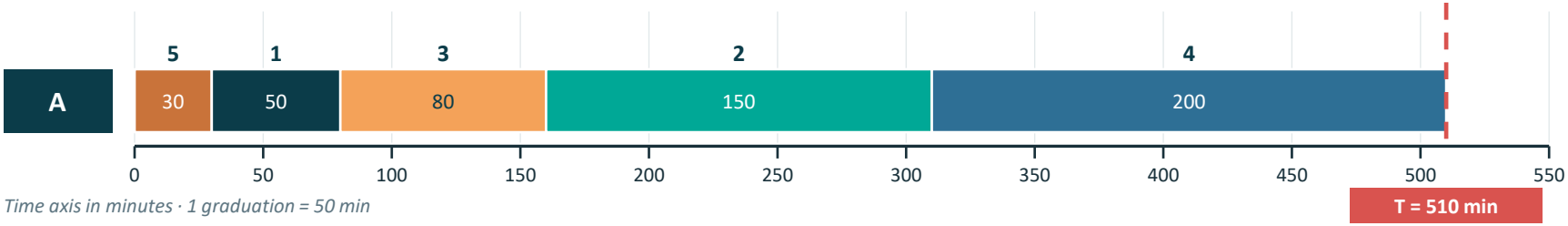
Example: 3 jobs on A then B — A: 2, 4, 2 min · B: 2, 3, 2 min



Time axis in minutes · 1 graduation = 1 min

1.2 EXAMPLE — GANTT CHART

MOT sequence on machine A



Time axis in minutes · 1 graduation = 50 min

Job	5	1	3	2	4
Cj (min)	30	80	160	310	510

$\bar{C} = 218$ min

NEXT → *Practise the MOT rule on a real case — Exercise 1.*

EXERCISE 1 · ONE MACHINE

A 3D printer in a fab lab

A university fab lab has a single 3D printer. This morning six print jobs were submitted, in the order 1 to 6. Printing times are in minutes.

Print job	1	2	3	4	5	6	Total
Ti (min)	45	120	30	90	60	15	**360**

QUESTIONS

1. Determine the optimal sequence with the MOT rule.
2. Draw the Gantt chart.
3. Determine the minimum mean completion time and compare it with FIFO (order of arrival).

EXERCISE 1 · SOLUTION

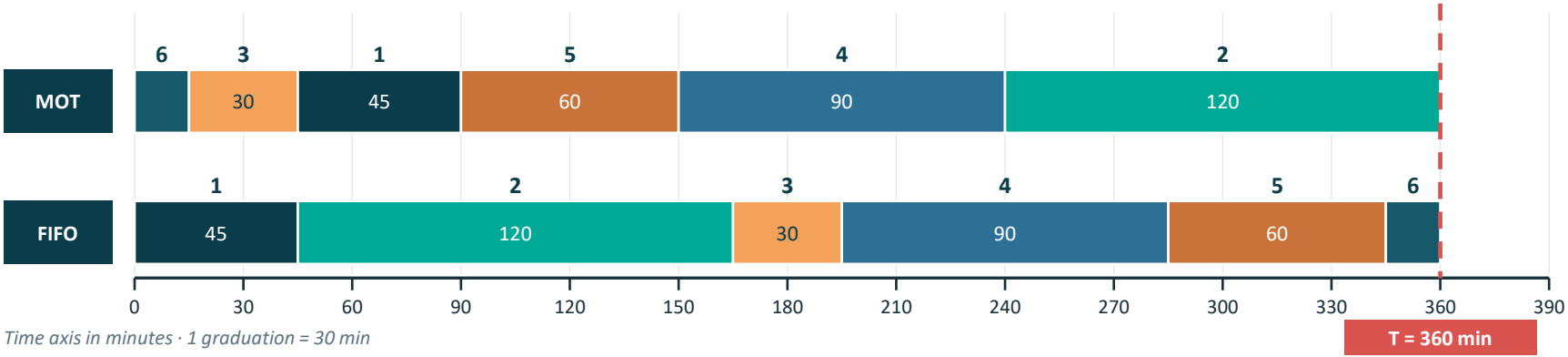
MOT sequence and completion times

Position j	1	2	3	4	5	6	Total
Job (MOT)	6	3	1	5	4	2	—
Tj (min)	15	30	45	60	90	120	**360**
Cj (min)	15	45	90	150	240	360	**900**

1	Sort by increasing time	*15 < 30 < 45 < 60 < 90 < 120*	**6–3–1–5–4–2**
2	Mean completion time (MOT)	*C = 900 / 6*	**150 min**
3	Mean completion time (FIFO)	*(45+165+195+285+345+360) / 6*	**232.5 min**

EXERCISE 1 · SOLUTION

Gantt charts — MOT versus FIFO



Time axis in minutes · 1 graduation = 30 min

Same total (360 min) — but MOT: C = 150 min vs FIFO: C = 232.5 min.

NEXT → *With two machines, the total time itself depends on the order — Section 2.*

2. SCHEDULING n JOBS ON TWO MACHINES

Two situations to distinguish

Each job needs two machines, A and B. TiA and TiB are the operating times of job i on A and on B.

CASE 1

Same routing for all jobs

Every job goes on A, then on B.

→ Johnson's algorithm (1954)

CASE 2

Different routings

Some jobs go A→B, some B→A, some use one machine only.

→ Jackson's algorithm (1957)

Criterion: minimise the total execution time (makespan), including the idle times Z of the machines.

2.1 ALL JOBS ON A, THEN ON B

Example — five jobs on two machines

Five jobs must be processed successively on machine A, then on machine B. Times in minutes.

Job i	1	2	3	4	5	Total
Ti (A)	50	150	80	200	30	**510**
Ti (B)	60	50	150	70	200	**530**

QUESTIONS

1. Determine the optimal sequence (Johnson's algorithm).
2. Draw the Gantt chart.
3. Determine the minimum total execution time.

2.1 JOHNSON'S ALGORITHM (1954)

The intuition behind the rule

Observation 1 — at the start

While the first job is on A, machine B is idle (Z).
→ Put FIRST the job with the shortest time on A.

Observation 2 — at the end

While the last job is on B, machine A is idle (Z).
→ Put LAST the job with the shortest time on B.



Keep these two Z as small as possible.

2.1 JOHNSON'S ALGORITHM

The three steps

01 Find the smallest operating time among ALL the remaining times (on A and on B).

02 If it is on machine A → place the job in the FIRST free position.
If it is on machine B → place the job in the LAST free position.

03 Cross out this job and go back to step 1 until all jobs are placed.

Fill the sequence from both ends towards the middle:



2.1 EXAMPLE — SOLUTION

Step 1 — applying Johnson's rule

Iteration	Smallest time	Job / machine	Decision	Sequence
1	30	Job 5 on A	front → position 1	5 _ _ _ _
2	50	Job 1 on A	front → position 2	5 1 _ _ _
3	50	Job 2 on B	back → position 5	5 1 _ _ 2
4	70	Job 4 on B	back → position 4	5 1 _ 4 2
5	**80**	**Job 3 (last one)**	**remaining → position 3**	**5 1 3 4 2**

JOHNSON SEQUENCE **5 – 1 – 3 – 4 – 2**

2.1 EXAMPLE — SOLUTION

Step 2 — the table in Johnson order

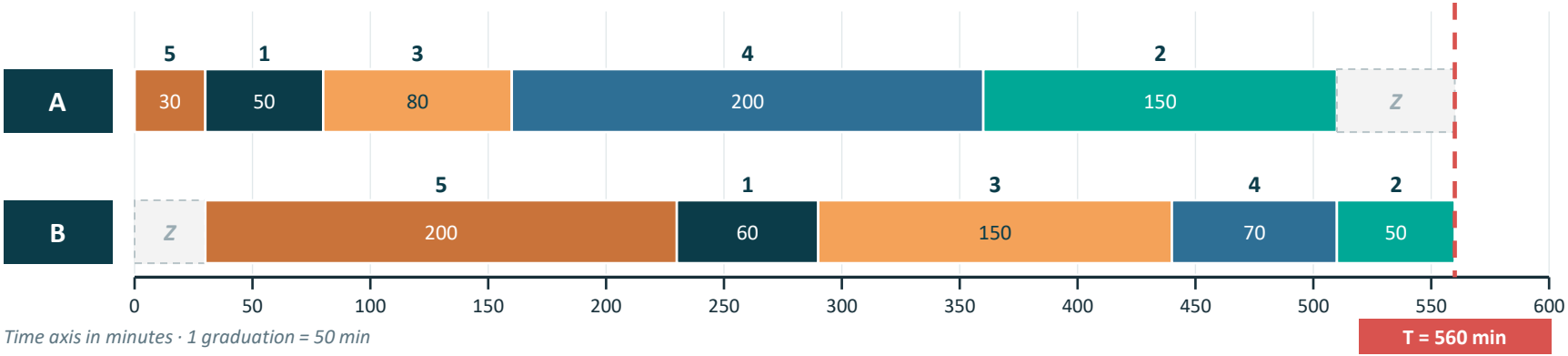
Job (in order)	5	1	3	4	2
Ti (A)	30	50	80	200	150
Ti (B)	200	60	150	70	50
End on A	30	80	160	360	510
End on B	**230**	**290**	**440**	**510**	**560**

Start on B = the LATER of (end of the job on A) and (end of the previous job on B).

Last value of the row "End on B" = total execution time: T = 560 min.

2.1 EXAMPLE — SOLUTION

Step 3 — Gantt chart and minimum time



Time axis in minutes · 1 graduation = 50 min

Minimum total execution time: T = 560 min. Machine B is idle only 30 min, at the start.

EXERCISE 2 · JOHNSON

A smartphone repair shop

Every phone goes first to diagnosis (station A), then to repair (station B). Six phones are waiting this morning. Times in minutes.

Phone	1	2	3	4	5	6	Total
Diagnosis (A)	25	10	40	15	35	20	**145**
Repair (B)	30	45	20	10	50	25	**180**

QUESTIONS

1. Determine the optimal sequence of the phones.
2. Draw the Gantt chart.
3. Determine the minimum total time and compare it with the order of arrival 1-2-3-4-5-6.

EXERCISE 2 · SOLUTION

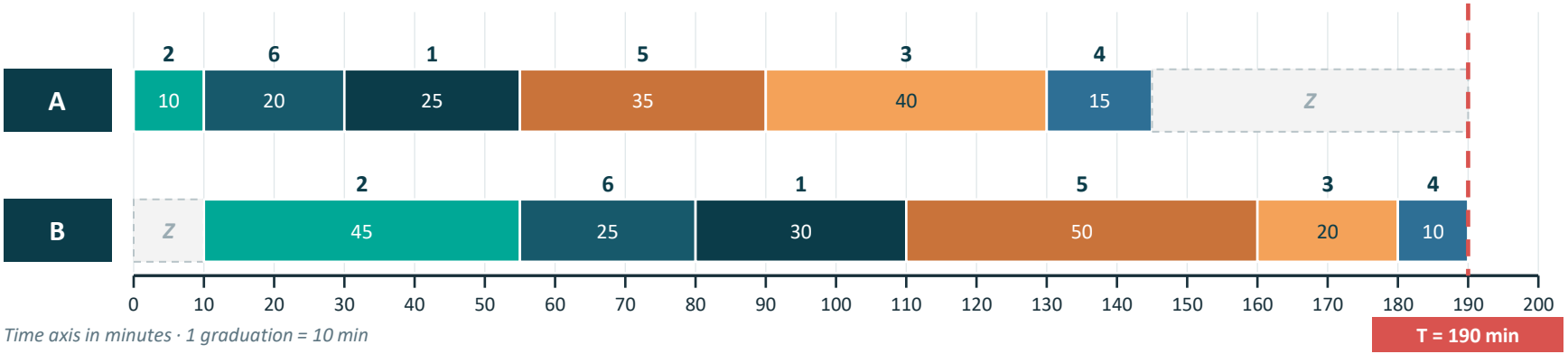
Johnson sequence and order table

Iteration	Smallest time	Job / machine	Position
1	10	Phone 2 on A · Phone 4 on B	pos. 1 · pos. 6
2	20	Phone 6 on A · Phone 3 on B	pos. 2 · pos. 5
3	25	Phone 1 on A	pos. 3
4	—	Phone 5 (remaining)	pos. 4

Job (in order)	2	6	1	5	3	4
Diagnosis (A)	10	20	25	35	40	15
Repair (B)	45	25	30	50	20	10
End on A	10	30	55	90	130	145
End on B	**55**	**80**	**110**	**160**	**180**	**190**

EXERCISE 2 · SOLUTION

Gantt chart — T = 190 min



Time axis in minutes · 1 graduation = 10 min

Optimal: 190 min · order of arrival: 205 min → 15 min saved.

NEXT → *When jobs do NOT all follow the same route → Jackson's algorithm.*

2.2 JOBS WITH DIFFERENT ROUTINGS

Example — Jackson's algorithm (1957)

Table 1 — routing A → B (0 = the job does not use this machine)

Job i	1	2	3	4	5	6	Total
Ti (A) first	50	80	10	50	30	70	**290**
Ti (B) then	30	60	30	0	0	0	**120**

Table 2 — routing B → A

Job i	7	8	9	10	11	Total
Ti (B) first	9	20	10	40	10	**89**
Ti (A) then	7	30	100	0	0	**137**

Workload — A: 427 min · B: 209 min

1. **Determine the optimal schedule.**
2. **Draw the Gantt chart.**
3. **Determine the minimum total time.**

2.2 JACKSON — STEP 1

Split the jobs into four subsets

AB

A then B

Jobs 1, 2, 3

Job	1	2	3
A	50	80	10
B	30	60	30

BA

B then A

Jobs 7, 8, 9

Job	7	8	9
B	9	20	10
A	7	30	100

A

A only

Jobs 4, 5, 6

Job	4	5	6
A	50	30	70

B

B only

Jobs 10, 11

Job	10	11
B	40	10

2.2 JACKSON — STEP 2

Best order inside each subset

AB	**Johnson on (A, B)** *min 10: job 3 on A → front · min 30: job 1 on B → back*	**3 – 2 – 1**
BA	**Johnson on (B, A) — B is the first machine** *min 7: job 7 on A (2nd) → back · min 10: job 9 on B (1st) → front*	**9 – 8 – 7**
A	**One machine → MOT rule** *30 < 50 < 70*	**5 – 4 – 6**
B	**One machine → MOT rule** *10 < 40*	**11 – 10**

2.2 JACKSON — STEP 3

Combining the subsets on each machine

MACHINE A	**AB** sent to B early	→	**A** fills the middle	→	**BA** received from B
MACHINE B	**BA** sent to A early	→	**B** fills the middle	→	**AB** received from A

Why? Each machine STARTS with the jobs it must send to the other machine, so the other machine is fed quickly, and ENDS with the jobs it receives from the other machine, which are ready long before. Single-machine jobs fill the middle.

Inside each block, keep the order found in step 2.

2.2 JACKSON — STEP 3 APPLIED

Sequences and times on each machine

Machine A : AB (3, 2, 1) → A (5, 4, 6) → BA (9, 8, 7)

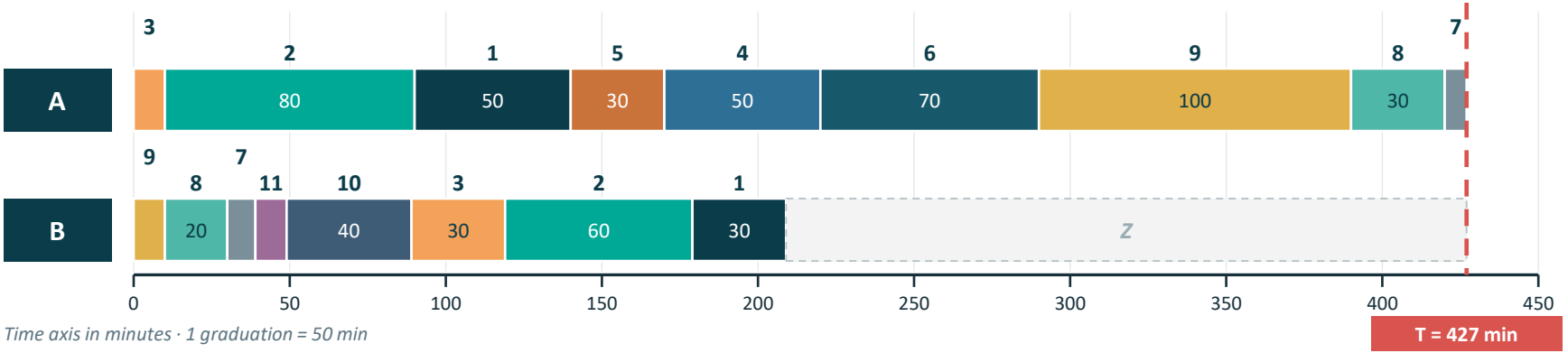
Job	3	2	1	5	4	6	9	8	7
Group	AB	AB	AB	A	A	A	BA	BA	BA
Ti (A)	10	80	50	30	50	70	100	30	7
End on A	**10**	**90**	**140**	**170**	**220**	**290**	**390**	**420**	**427**

Machine B : BA (9, 8, 7) → B (11, 10) → AB (3, 2, 1)

Job	9	8	7	11	10	3	2	1
Group	BA	BA	BA	B	B	AB	AB	AB
Ti (B)	10	20	9	10	40	30	60	30
End on B	**10**	**30**	**39**	**49**	**89**	**119**	**179**	**209**

2.2 JACKSON — SOLUTION

Gantt chart and minimum time



Time axis in minutes · 1 graduation = 50 min

Tmin = 427 min = total workload of machine A, which never stops → this schedule is optimal.

EXERCISE 3 · JACKSON

A car service garage

Bay A = mechanics, bay B = electronic diagnosis. Eight cars, times in minutes (— = not needed).

Car	1	2	3	4	5	6	7	8	Total
Routing	A→B	A→B	A→B	B→A	B→A	A	A	B	—
Ti (A)	40	25	55	35	20	30	45	—	**250**
Ti (B)	20	35	15	15	30	—	—	25	**140**

QUESTIONS

1. Determine the optimal schedule (Jackson's algorithm).
2. Draw the Gantt chart.
3. Determine the minimum total time.

EXERCISE 3 · SOLUTION

Steps 1, 2 and 3

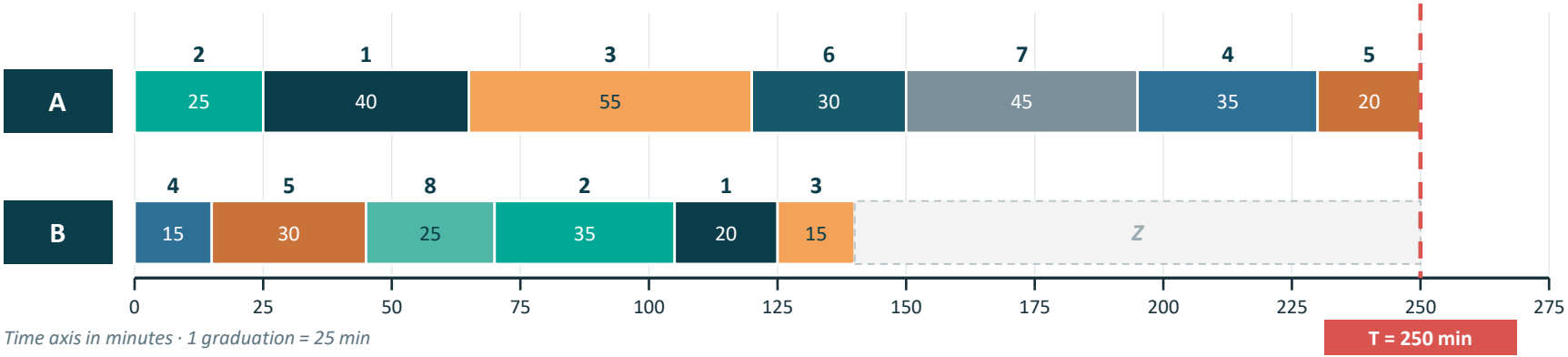
Subset	Jobs	Method	Best order
AB	1, 2, 3	Johnson on (A, B): 15 car 3 on B → back · 20 car 1 on B → back	2 – 1 – 3
BA	4, 5	Johnson on (B, A): 15 car 4 on B (1st machine) → front	4 – 5
A only	6, 7	MOT: 30 < 45	6 – 7
B only	8	single job	8

Bay A: AB → A → BA	2	1	3	6	7	4	5
End on A	25	65	120	150	195	230	250

Bay B: BA → B → AB	4	5	8	2	1	3
End on B	15	45	70	105	125	140

EXERCISE 3 · SOLUTION

Gantt chart — T = 250 min



Time axis in minutes · 1 graduation = 25 min

T = 250 min = workload of bay A, which never stops → optimal.

NEXT → *Three machines in series → reduce the problem to two machines — Section 3.*

3. SCHEDULING n JOBS ON THREE MACHINES

Reducing three machines to two

Johnson's algorithm only applies to two machines. The three-machine case can be brought back to two machines if the middle machine B is dominated by A or by C.

Dominance by A

$$\min Ti(A) \geq \max Ti(B)$$

or dominance by C

$$\min Ti(C) \geq \max Ti(B)$$

If ONE of the two conditions holds, build two fictitious machines and apply Johnson:

Fictitious machine 1	$G = A + B$
Fictitious machine 2	$H = B + C$

3. EXAMPLE

Seven jobs on three machines

Each job goes on A, then B, then C. Times in minutes.

Job i	1	2	3	4	5	6	7	Total
Ti (A)	20	12	19	16	14	12	17	**110**
Ti (B)	4	1	9	12	5	7	8	**46**
Ti (C)	7	11	4	18	18	3	6	**67**

QUESTIONS

1. Determine the optimal sequence.
2. Draw the Gantt chart.
3. Determine the minimum total execution time.

3. EXAMPLE — SOLUTION

Steps 1 and 2 — dominance and fictitious machines

1	**Dominance by A?**	*min Ti(A) = 12 ≥ max Ti(B) = 12*	**TRUE ✓**
2	**Dominance by C?**	*min Ti(C) = 3 < max Ti(B) = 12*	**false (not needed)**

Fictitious machines (with sum of times)

Job i	1	2	3	4	5	6	7	Total
G = A + B	24	13	28	28	19	19	25	**156**
H = B + C	11	12	13	30	23	10	14	**113**

3. EXAMPLE — SOLUTION

Step 3 — Johnson on G and H, then real times

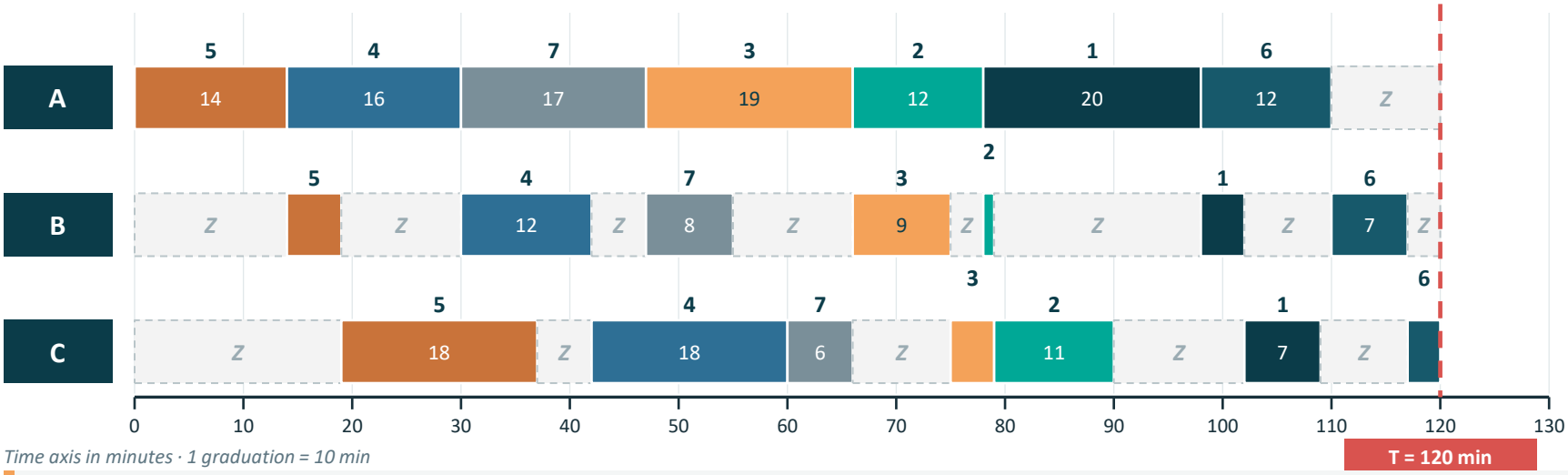
Smallest	10	11	12	13	14	19	—
Job / machine	6 on H	1 on H	2 on H	3 on H	7 on H	5 on G	4
Position	7	6	5	4	3	1	2

Optimal sequence: 5 – 4 – 7 – 3 – 2 – 1 – 6

Job (in order)	5	4	7	3	2	1	6
Ti (A)	14	16	17	19	12	20	12
Ti (B)	5	12	8	9	1	4	7
Ti (C)	18	18	6	4	11	7	3
End on A	14	30	47	66	78	98	110
End on B	19	42	55	75	79	102	117
End on C	**37**	**60**	**66**	**79**	**90**	**109**	**120**

3. EXAMPLE — SOLUTION

Gantt chart on the real machines A, B, C



Time axis in minutes · 1 graduation = 10 min

Minimum total execution time: T = 120 min.

3. LIMITS OF THE METHOD

When the dominance condition fails

Condition TRUE

Reduction to G = A + B and H = B + C is valid → Johnson gives the optimal order.

Condition FALSE

The reduction can still be used, but only as a heuristic: good order, no guarantee of optimality.

Exact solution: model the problem as an integer linear programme and solve it by branch and bound (e.g. Excel Solver, Python OR-Tools) — beyond the scope of this course.

NEXT → *Practise the three-machine method — Exercise 4.*

EXERCISE 4 · THREE MACHINES

An e-commerce warehouse

Each online order goes through picking (A), quality check (B) and packing (C). Six orders, times in minutes.

Order	1	2	3	4	5	6	Total
Picking (A)	12	8	15	10	9	14	**68**
Check (B)	3	5	2	6	4	3	**23**
Packing (C)	10	7	6	12	11	8	**54**

QUESTIONS

1. Check the dominance condition and determine the optimal sequence.
2. Draw the Gantt chart.
3. Determine the minimum total time and compare with the order 1-2-3-4-5-6.

EXERCISE 4 · SOLUTION

Dominance, fictitious machines, Johnson

Step 1 — Dominance by A: min Ti(A) = 8 ≥ max Ti(B) = 6 → TRUE ✓

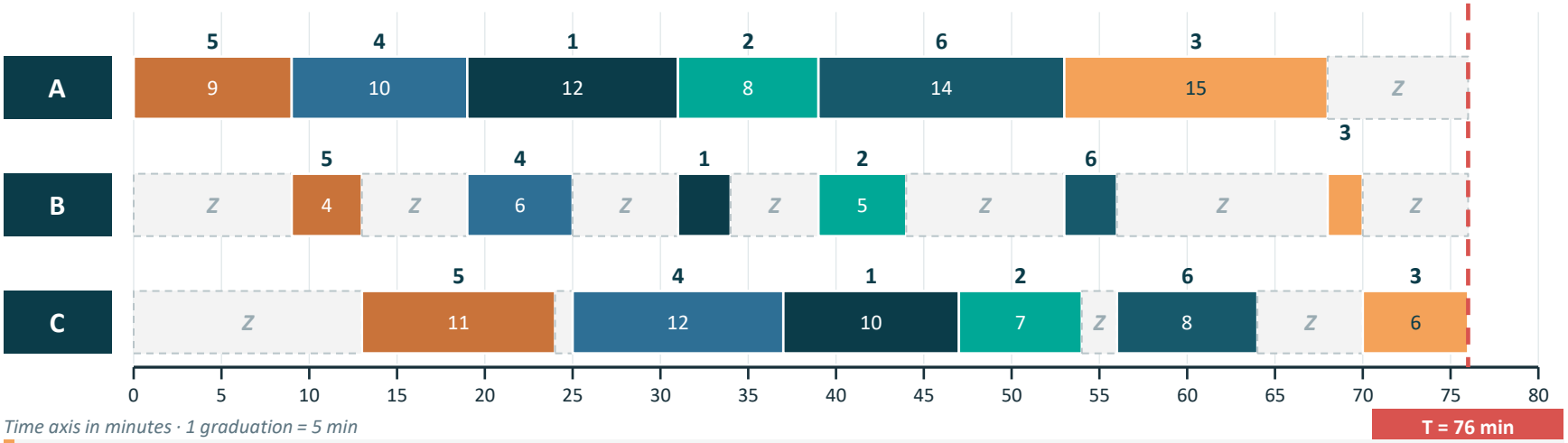
Order	1	2	3	4	5	6	Total
G = A + B	15	13	17	16	13	17	**91**
H = B + C	13	12	8	18	15	11	**77**

Johnson on G, H: 8 (3 on H) · 11 (6 on H) · 12 (2 on H) · 13 (5 on G, 1 on H) · 16 (4 on G) → 5 – 4 – 1 – 2 – 6 – 3

Job (in order)	5	4	1	2	6	3
Ti (A)	9	10	12	8	14	15
Ti (B)	4	6	3	5	3	2
Ti (C)	11	12	10	7	8	6
End on A	9	19	31	39	53	68
End on B	13	25	34	44	56	70
End on C	**24**	**37**	**47**	**54**	**64**	**76**

EXERCISE 4 · SOLUTION

Gantt chart — T = 76 min



Time axis in minutes · 1 graduation = 5 min

Optimal: 76 min · order of arrival: 82 min → 6 min saved.

NEXT → *More than three machines → heuristic methods — Section 4.*

4. SCHEDULING n JOBS ON m MACHINES

From optimal solution to heuristic

NUMBER OF POSSIBLE SCHEDULES

$$(n!)^m$$

6 jobs, 4 machines: $(720)^4 \approx 2.7 \times 10^{11}$

A heuristic method

Gives a good — often near-optimal — solution, but without a guarantee of optimality.

The CDS heuristic (Campbell, Dudek & Smith) — same routing for all jobs:

1. Group the machines into m − 1 fictitious two-machine problems
2. Solve each problem with Johnson's algorithm
3. Evaluate each sequence on the real machines (Gantt) and keep the best

4. LARGE PROBLEMS

What to do with a large number of machines?

Approach	Examples	When / why
Other heuristics	NEH (Nawaz–Enscore–Ham), Palmer slope index	Very good quick solutions for flow shops
Priority (dispatching) rules	SPT, EDD, CR… applied machine by machine	Simple; used daily in real job-shops
Bottleneck approach	Schedule the bottleneck first (Theory of Constraints)	Focus where capacity is lost
Metaheuristics	Genetic algorithms, simulated annealing, tabu search	Near-optimal for large problems
Exact methods	Integer programming + branch & bound (Gurobi, CPLEX, OR-Tools)	Optimal, for medium-size problems
Software & simulation	ERP / APS / MES scheduling modules, simulation	Industrial practice, what-if scenarios

4. EXAMPLE

Six jobs on four machines

Each job goes on A, B, C, then D. Times in minutes.

Job i	1	2	3	4	5	6	Total
Ti (A)	5	8	7	8	6	1	**35**
Ti (B)	4	9	4	6	1	8	**32**
Ti (C)	1	9	2	1	6	6	**25**
Ti (D)	4	7	9	9	1	7	**37**

QUESTIONS

1. Determine the best sequence with the heuristic.
2. Draw the Gantt charts.
3. Determine the minimum total time.

4. EXAMPLE — SOLUTION

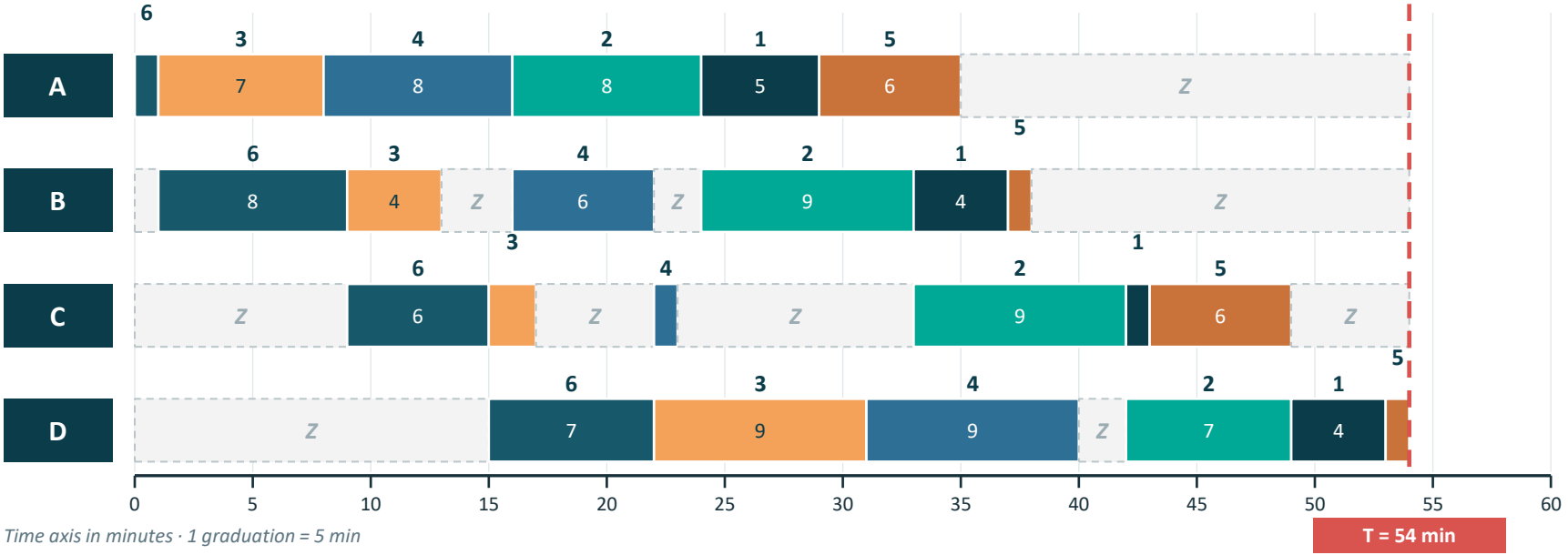
Step 1 — three fictitious problems

Problem	Fictitious machine	1	2	3	4	5	6	Johnson sequence
P1 A | D	G (1st)	5	8	7	8	6	1	6–3–4–2–1–5
	H (2nd)	4	7	9	9	1	7	
P2 A+B | C+D	G (1st)	9	17	11	14	7	9	5–6–3–2–4–1
	H (2nd)	5	16	11	10	7	13	
P3 A+B+C | B+C+D	G (1st)	10	26	13	15	13	15	3–6–4–2–1–5
	H (2nd)	9	25	15	16	8	21	

P3: tie at 15 (jobs 4 and 6 on G) — order 6 then 4 retained. Each candidate must now be tested on the real machines.

4. EXAMPLE — SOLUTION

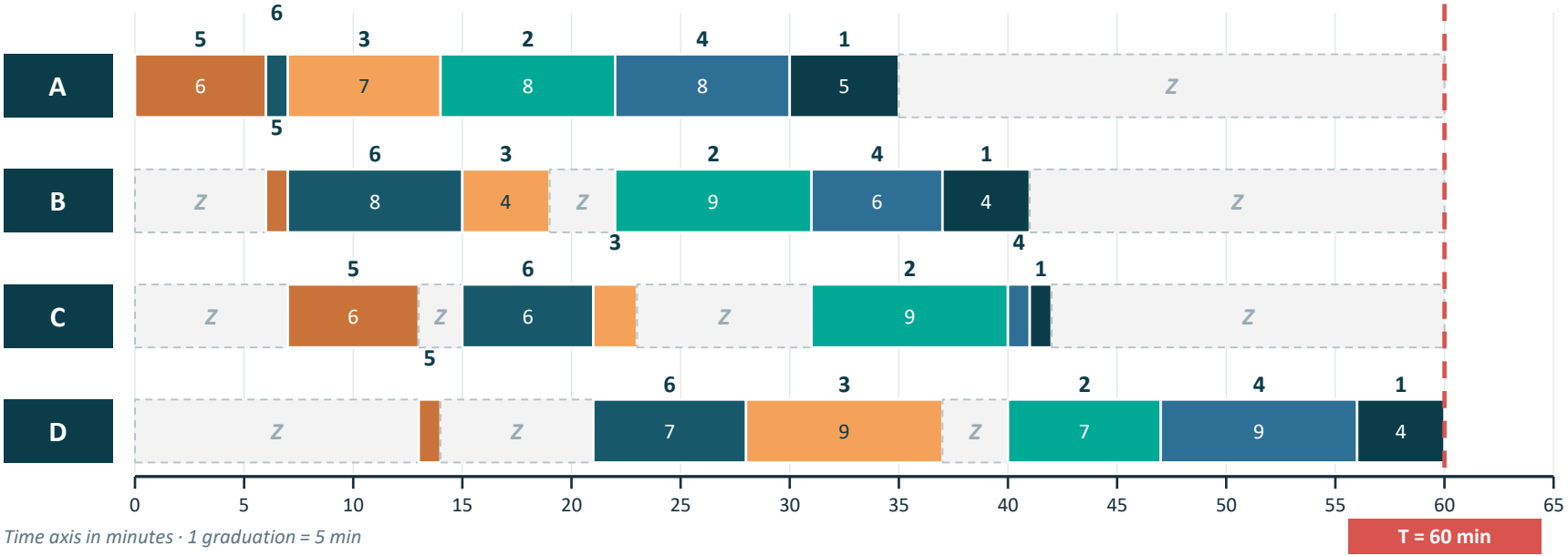
Gantt chart — problem 1: 6 – 3 – 4 – 2 – 1 – 5



Time axis in minutes · 1 graduation = 5 min

4. EXAMPLE — SOLUTION

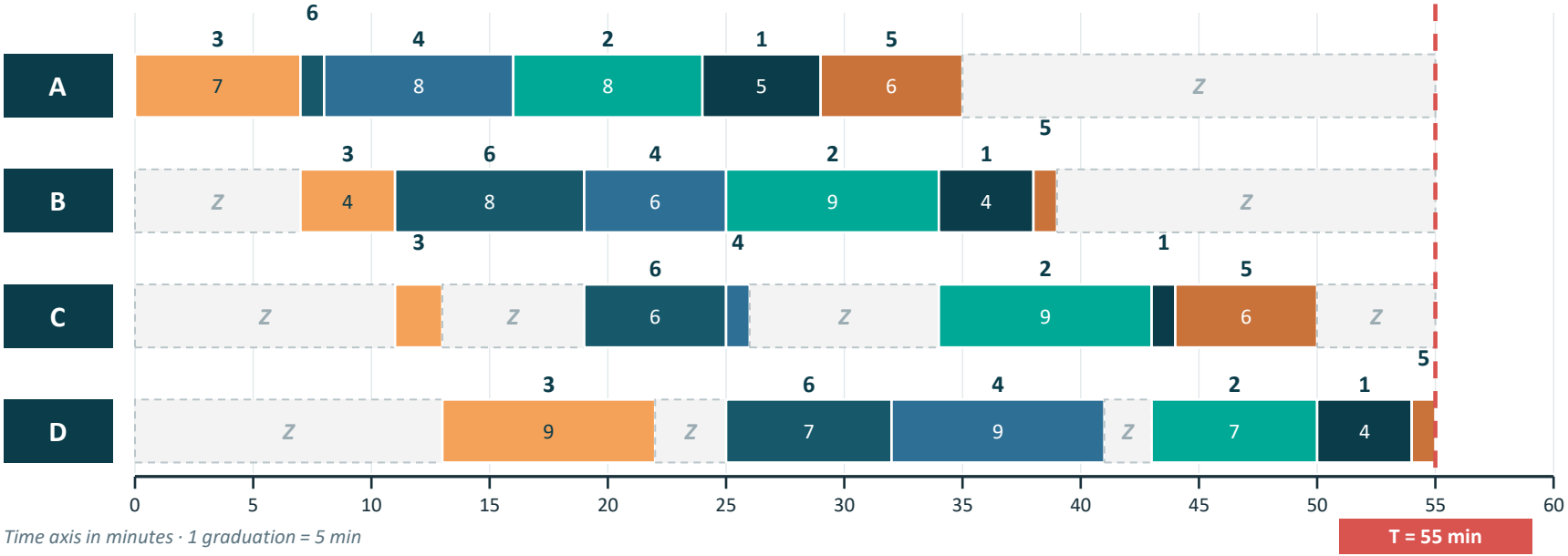
Gantt chart — problem 2: 5 – 6 – 3 – 2 – 4 – 1



Time axis in minutes · 1 graduation = 5 min

4. EXAMPLE — SOLUTION

Gantt chart — problem 3: 3 – 6 – 4 – 2 – 1 – 5



Time axis in minutes · 1 graduation = 5 min

4. EXAMPLE — SOLUTION

Comparing the candidates

Problem 1

6 – 3 – 4 – 2 – 1 – 5

T = 54

Problem 2

5 – 6 – 3 – 2 – 4 – 1

T = 60

Problem 3

3 – 6 – 4 – 2 – 1 – 5

T = 55

Machine	A	B	C	D
Workload (min)	35	32	25	37
Utilisation (T = 54)	65 %	59 %	46 %	69 %

Retained: 6 – 3 – 4 – 2 – 1 – 5, T = 54 min (also the true optimum, found by full enumeration).

EXERCISE 5 · FOUR MACHINES

A generic-medicine production line

Each batch goes through weighing (A), mixing (B), tableting (C) and packaging (D). Five batches, times in minutes.

Batch	1	2	3	4	5	Total
Weighing (A)	6	3	8	5	4	**26**
Mixing (B)	4	6	3	7	5	**25**
Tableting (C)	5	4	6	3	7	**25**
Packaging (D)	3	8	5	6	9	**31**

QUESTIONS

1. Build the three fictitious problems and apply Johnson to each.
2. Draw the Gantt chart of each candidate and determine the minimum total time.

EXERCISE 5 · SOLUTION

Three fictitious problems and their results

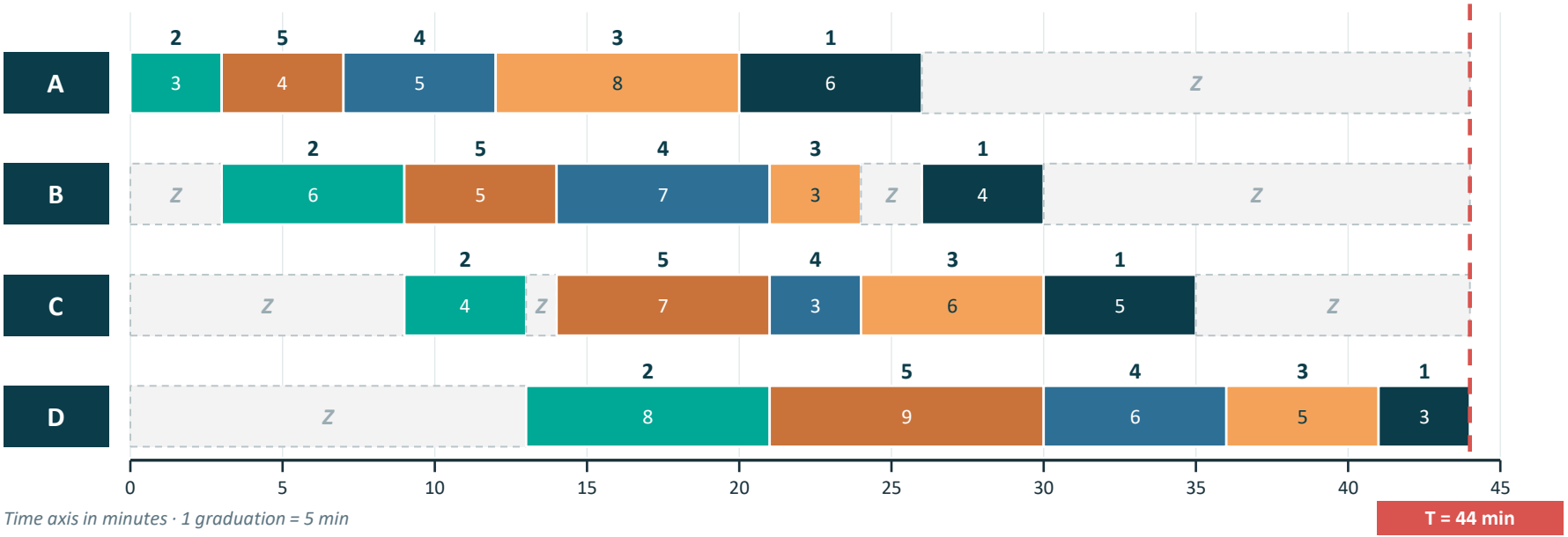
Problem	1	2	3	4	5	Sequence	T (min)
P1: A | D — G	6	3	8	5	4	2–5–4–3–1	44
H	3	8	5	6	9		
P2: A+B | C+D — G	10	9	11	12	9	2–5–3–4–1	44
H	8	12	11	9	16		
P3: A+B+C | B+C+D — G	15	13	17	15	16	2–4–5–3–1	45
H	12	18	14	16	21		

Best total: 44 min (P1 and P2) — order of arrival 1-2-3-4-5: 50 min.

Ties: in P1, value 3 appears on A (batch 2 → front) and on D (batch 1 → back).

EXERCISE 5 · SOLUTION

Gantt chart — sequence 2 – 5 – 4 – 3 – 1



Time axis in minutes · 1 graduation = 5 min

CHAPTER SUMMARY

Which method for which case?

Situation	Method	Criterion	Result
1 machine	MOT / SPT rule	Mean completion time C	Optimal
2 machines, same routing	Johnson	Makespan T	Optimal
2 machines, different routings	Jackson (AB,A,BA | BA,B,AB)	Makespan T	Optimal
3 machines	Dominance + Johnson on A+B, B+C	Makespan T	Optimal if dominance holds
m machines	CDS heuristic (m−1 Johnson problems)	Makespan T	Good, not guaranteed

> ***In every case: draw the Gantt chart, mark the idle times Z, read the minimum time.***

FINAL PRACTICE

Two quick exercises

Exercise A — one machine (min)

Job	1	2	3	4	5	Total
Ti (min)	6	2	8	3	9	**28**

1. Optimal MOT sequence 2. Gantt chart 3. Minimum mean completion time

Exercise B — two machines, A then B (min)

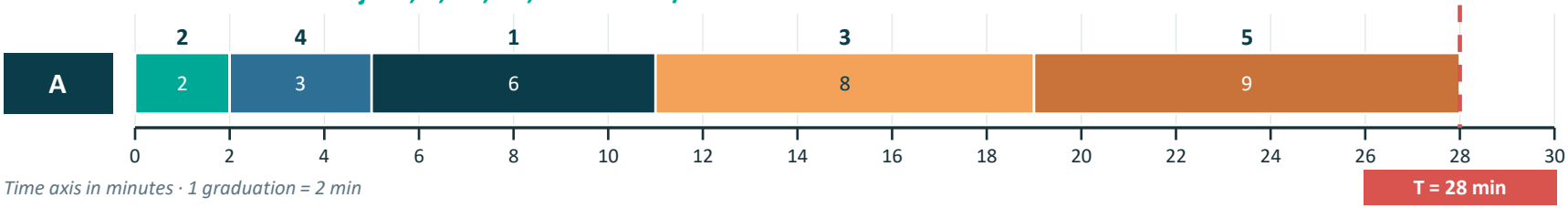
Job	1	2	3	4	5	Total
Ti (A)	5	1	9	3	10	**28**
Ti (B)	2	6	7	8	4	**27**

1. Optimal sequence (Johnson) 2. Gantt chart 3. Minimum total time

FINAL PRACTICE · SOLUTIONS

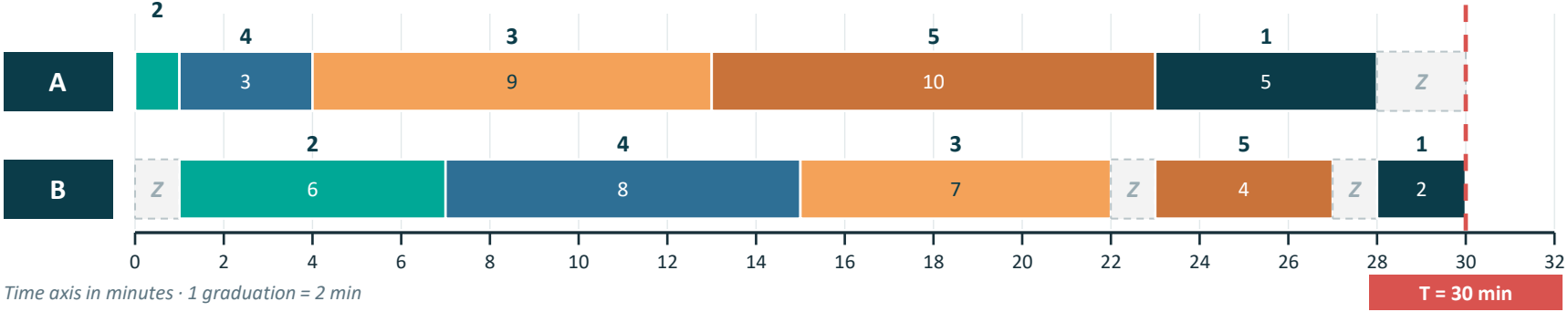
A: C̄ = 13 min · B: T = 30 min

A — MOT: 2 – 4 – 1 – 3 – 5 · Cj = 2, 5, 11, 19, 28 · C̄ = 65 / 5 = 13 min



Time axis in minutes · 1 graduation = 2 min

B — Johnson: 2 – 4 – 3 – 5 – 1



Time axis in minutes · 1 graduation = 2 min

Thank you for your attention

Next session: tutorial — bring squared paper and a ruler for the Gantt charts.

MOT rule | **Johnson** | **Jackson** | **Heuristic**

Dr Miryam Daoud Marrakchi · Tunis Business School · Production & Operations Management · October 2026